Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build a Multi AI Agents Workflow for Stock Trading (C177)

Course Code: C177

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	25 October 2026	New course: Build a Multi AI Agents Workflow for Stock Trading - eight agents (lead, finance research, news sentiment, trading strategy, signal generation, stock trader, performance analysis, strategy review), built with AI vibe coding over eight labs; 9:30am - 5:30pm, 7.5 hours.	Dr Alfred Ang

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Course Information

Course Title	Build a Multi AI Agents Workflow for Stock Trading (C177)
Course Reference	C177
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 7.5 instructional hours per day (7.5 hours)
Daily Timing	9:30 am – 5:30 pm (7.5 instructional hours; 30-minute lunch)
Mode	Instructor-led, hands-on AI vibe coding labs
Prerequisite	AI Powered Quantitative Trading Fundamentals (C728) - the course that demonstrates this multi AI agents trading system; C177 teaches you to build it from scratch
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Design a multi AI agents trading workflow: the eight agent roles, the message contract between them and a lead agent that orchestrates the flow.
- LO2: Vibe-code the research agents that gather Twelve Data candles, Yahoo Finance and Equibles company data, and Reddit and NewsAPI sentiment.
- LO3: Vibe-code the decision and execution agents that formulate a strategy from trend, momentum, volatility, VWAP, RSI and MACD, generate long, short or null signals and place orders on an Alpaca paper account.
- LO4: Close the loop with a performance dashboard and a strategy review agent, orchestrate the full workflow and publish the project to GitHub.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — From Design to a Self-Improving Multi AI Agents Trading Desk

Time	Duration	Topic / Activity
9:30–9:45	15 min	Welcome, course outcomes, safety boundary (paper trading only) and set-up check
9:45–10:15	30 min	**Topic 1 - the eight agents,**

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		the message contract, lead orchestration and the AI vibe coding loop
10:15–10:50	35 min	Hands-on: Lab 1: Scaffold the Project and the Agent Message Contract
10:50–11:10	20 min	**Topic 2 - multi-timeframe candles, company fundamentals and Equibles data**
11:10–11:45	35 min	Hands-on: Lab 2: Build the Finance Research Agent
11:45–12:00	15 min	**Topic 2 - sentiment from Reddit and news, Reddit OAuth**
12:00–12:35	35 min	Hands-on: Lab 3: Build the News Sentiment Agent with Reddit and NewsAPI
12:35–13:05	30 min	Lunch break
13:05–13:20	15 min	**Topic 2 - trend, momentum, volatility, VWAP, RSI and MACD as votes; mixing algorithms**
13:20–13:55	35 min	Hands-on: Lab 4: Build the Trading Strategy Agent
13:55–14:05	10 min	**Topic 2 - from a strategy score to a long, short or null signal**
14:05–14:25	20 min	Hands-on: Lab 5: Build the Signal Generation Agent
14:25–14:40	15 min	**Topic 3 - automated execution on Alpaca paper: long, short and flipping**
14:40–15:10	30 min	Hands-on: Lab 6: Build the Stock Trader Agent on Alpaca Paper
15:10–15:25	15 min	**Topic 3 - return, Sharpe, drawdown and win rate; building a dashboard**
15:25–15:55	30 min	Hands-on: Lab 7: Build the Performance Analysis Agent and the Dashboard
15:55–16:10	15 min	**Topic 3 - feedback loops, lead orchestration and publishing to GitHub**
16:10–17:15	65 min	Hands-on: Lab 8: Close the Loop: Strategy Review, Lead Agent and GitHub
17:15–17:30	15 min	Run the full workflow, recap, next steps and Q&A

Total scheduled time: 450 minutes (7 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 0 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs

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Topic 01: Designing a Multi AI Agents Trading Workflow	—	Lab 1
Topic 02: Building the Research and Decision Agents	—	Lab 2, Lab 3, Lab 4, Lab 5
Topic 03: Execution, Performance and the Feedback Loop	—	Lab 6, Lab 7, Lab 8

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